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Research Article

Programmable dielectric loss in gradient carbon nanostructures with built-in electric field boosting dual-band electromagnetic wave absorption

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ABSTRACT

Carbon nanotubes are promising electromagnetic wave (EMW) absorption materials due to their excellent conduction loss and multiple polarization relaxations. However, their high electrical conductivity often causes severe impedance mismatch and small skin depth, leading to strong reflection rather than efficient absorption. Herein, a gradient carbon engineering strategy is proposed to enhance dielectric loss for broadband EMW absorption. Density functional theory calculations confirm that built-in electric fields (BIEFs) synergistically couple with the gradient carbon nanostructure to promote polarization relaxation. Notably, BIEFs facilitate electron transfer and modulate charge distribution, thereby substantially boosting interfacial polarization. With balanced dielectric loss and impedance matching, the resultant composite exhibits dual-band EMW absorption, achieving a minimum reflection loss of −42.3 dB at 5.3 GHz and an effective absorption bandwidth of 3.1 GHz across the C and Ku bands. In addition, the polydimethylsiloxane–blended composite also shows good thermal conductivity. This work demonstrates the effectiveness of gradient carbon engineering in tailoring dielectric loss and provides a new strategy for designing high-performance EMW absorption materials via BIEFs.

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1. Introduction

The rapid proliferation of 5G, wireless devices, and radar technology has created a complex electromagnetic (EM) environment with multi–band electromagnetic waves (EMWs), posing escalating threats to human health and military security [1–4]. Broadband EMW–absorbing materials that efficiently convert EM energy into heat have therefore become a major research focus [5,6]. While synergistic dielectric-magnetic loss modulation enables broadband absorption beyond single-mechanism limits, the strong interdependence of EM parameters often disrupts this balance, making multiband impedance matching inherently challenging [7–9]. To address this, structurally tunable carbon–based materials are essential candidates for complex EM environments due to their controllable EM parameters, excellent dielectric loss, and structural flexibility.

Carbon nanotubes (CNTs) are important carbon materials for EMW attenuation due to their intrinsic conduction loss and polarization relaxation. However, their high conductivity (σ) commonly leads to impedance mismatch and small skin depth, favoring EMW reflection over absorption across diverse CNT systems [10–12]. Typically, raising CNT loading in polydimethylsiloxane (PDMS) from 0.5wt.% to 2wt.% reduced skin depth from 1.02 to 0.53 mm, leading to strong reflection for EM shielding [13]. A similar filler-content dependence was observed for multi-walled CNTs (1wt.%–5wt.%), and even single-walled CNTs reduced the skin depth to as low as 1–3 μm [14–16]. Nevertheless, effective gigahertz EMW absorption generally requires a matching thickness of at least 1 mm, as established by prior studies [17–19]. In addition to high σ, CNTs are limited by poor impedance matching due to their excessive permittivity, and their single loss mechanism restricts the absorption bandwidth, an issue shared by other carbon materials [20–22]. Therefore, rational design of CNT-based composites and microstructures to tailor σ and loss mechanisms is key to optimizing impedance matching and EMW absorption performance.

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Microscale surface engineering and macroscale composite construction offer feasible strategies to modulate the σ of CNT–based composites, enabling enhanced impedance matching and diversified loss mechanisms. In typical microscale surface engineering, CNTs are coated with conductive polymers, amorphous carbons, or metal oxides to optimize σ and electromagnetic parameters, achieving a synergistic balance between impedance matching and EMW attenuation [23–27]. To further advance the application of CNTs for EMW absorption, another strategy centers on macroscale composite construction, such as three-dimensional aerogels and porous foams [28–31]. These architectures can serve as physical barriers that induce multiple EMW reflection/scattering and as structural scaffolds that facilitate uniform CNT dispersion, thereby optimizing impedance matching and establishing an effective conductive network. Despite significant progress in microscale and macroscale structures, gradient architecture design has emerged as a research focus for EMW absorption. For instance, gradient graphene composites, magnetic-dielectric heterostructures, and magnetic domains show great potential for creating a gradient distribution of dielectric and magnetic components, thereby optimizing impedance matching and modulating EMW absorption mechanisms [32–34].

Herein, we propose a gradient carbon engineering strategy that encapsulates CNTs with an inner low-graphitized carbon layer and an outer amorphous carbon layer, enabling optimized impedance matching and superior EMW attenuation. Specifically, the low-graphitized carbon layer is controlled by metal-catalyzed graphitization, forming core-shell heterojunctions and graphitic domains. Owing to the synergistic regulation of σ and interface structure, a dual-peak EMW absorption spectrum was achieved in the C and Ku bands. By tuning the outer amorphous carbon layers, the minimum reflection loss (RL_{min}) and effective absorption bandwidth (EAB) were optimized to −42.3 dB at 5.3 and 3.1 GHz across the C and Ku bands, respectively. Density functional theory (DFT) analysis of the built-in electric fields (BIEFs) further elucidates the EMW absorption mechanism by confirming enhanced interfacial polarization, inspiring an innovative approach to constructing gradient carbon structures for dual–band or even multi–band broadband EMW absorption.

2. Results and discussion

2.1. Fabrication and structural characterization

The fabrication process of the gradient carbon nanostructures is illustrated in Fig. 1(a). Briefly, polymerization of a dianhydride with a diamine affords polyamide acid (PAA_1 and PAA_2). PAA_1 is then immobilized and assembled on CNTs to afford CNT–PI_1. Subsequent treatment with Fe/Co/Ni salts and hexamethylenetetramine (HMT) produces CNT–PI_1/FeCoNi trimetallic hydroxides (CNT–PI_1/FeCoNi THs), which become partially graphitized, FeCoNi–containing N–doped carbon ($NC_1{}^M$) on CNTs (CNT–$NC_1{}^M$) after pyrolysis. An additional amorphous carbon layer is finally introduced by a second PI assembly and pyrolysis, yielding CNT–$NC_1{}^M$–NC_1 and CNT–$NC_1{}^M$–NC_2. The surface microstructures were characterized by a scanning electron microscope (SEM). Unlike smooth CNTs and rough CNT-PI_1, CNT-PI_1/FeCoNi THs exhibit a rugged surface with nanoparticle protrusions (Fig. 1(b_1–b_3)). After pyrolysis, $NC_1{}^M$ encapsulates CNTs to form CNT-$NC_1{}^M$ (Fig. 1(b_4)), creating a CNT/$NC_1{}^M$ interface. Further PI assembly on CNT-$NC_1{}^M$ gives CNT-$NC_1{}^M$-PI_1 and CNT-$NC_1{}^M$-PI_2, showing an interconnected nanosheet-based morphology (Fig. 1(b_5 and b_7)), and this nanostructure is preserved in CNT-$NC_1{}^M$-NC_1 and CNT-$NC_1{}^M$-NC_2 (Fig. 1(b_6 and b_8)). Moreover, the uniformity and reproducibility of the gradient nanostructures across different batches are confirmed in Fig. S1 in the Supplementary Materials. Fourier-transform infrared (FT–IR) spectra of PI–containing intermediates (Fig. 1(c)) show stretching vibration of C=O at 1715 cm^{-1} and C–N bond at 1370 cm^{-1}, confirming successful PI incorporation [35]. X-ray diffraction (XRD) patterns of PI-containing intermediates and CNT show the existence of crystalline PI peaks at 17.6°–22.6° and graphitic carbon peak at ∼26° (Figs. 1(d) and S2). Fig. 1(e) illustrates the PI-derived gradient carbon engineering, where the graphitization degree and nitrogen doping are tailored to modulate conduction loss and polarization relaxation. This enables programmable dielectric loss in gradient carbon nanostructures, thereby enhancing EMW attenuation capability.

XRD patterns (Fig. 2(a)) of CNT–$NC_1{}^M$, CNT–$NC_1{}^M$–NC_1, and CNT–$NC_1{}^M$–NC_2 exhibit three peaks at ∼43.7°, 50.9°, and 74.8°, indexed to the (111), (200), and (220) planes with positions intermediate between those of pure Fe, Co, and Ni (PDF#89–4185, #15–0806, #01–1260), confirming the formation of FeCoNi solid-solution alloy. The peak at ∼26.5° corresponds to graphitic carbon in CNTs [36]. Raman spectra (Fig. S3) of the polyimide–derived carbons exhibit D and G bands at ∼1350 and ∼1580 cm^{-1}, corresponding to disordered and graphitic carbon, with their integrated area ratio (I_D/I_G) from two–peak fitting indicating the defect density [37]. The I_D/I_G ratio steadily increases from 0.10 (CNT) to 1.88 (CNT–$NC_1{}^M$), 2.37 (CNT–$NC_1{}^M$–NC_1), and 2.44 (CNT–$NC_1{}^M$–NC_2), demonstrating that gradient carbon engineering effectively modulates structural defects. To further analyze carbon defects, a four–peak Lorentzian deconvolution was performed (Fig. 2(b)), where the D_1 band corresponds to disordered carbon, the G band to in–plane stretching of sp^2-hybridized carbon in a graphitic lattice, and the D_2 and D_3 bands to disordered graphitic lattice and amorphous carbon, respectively [38]. Compared with CNT–$NC_1{}^M$ (I_{D1}/I_G=1.08), both CNT–$NC_1{}^M$–NC_1 (1.90) and CNT–$NC_1{}^M$–NC_2 (2.30) show higher ratios, indicating increased defect concentrations. Notably, CNT–$NC_1{}^M$–NC_1 and CNT–$NC_1{}^M$–NC_2 exhibit lower I_{D2}/I_G ratios (0.32 and 0.29) than CNT–$NC_1{}^M$ (0.46), but higher I_{D3}/I_G ratios (0.47 and 0.43 vs. 0.35). These trends (Fig. 2(c)) indicate that gradient carbon engineering reduces disordered graphitic lattice and increases amorphous carbon, enabling tunable defect structures and graphitization to tailor polarization relaxation and conduction loss. Magnetic properties of CNT-$NC_1{}^M$, CNT-$NC_1{}^M$-NC_1, and CNT-$NC_1{}^M$-NC_2 were determined from hysteresis loops (Fig. 2(d)), giving saturation magnetization (M_s) of 28.5, 14.9, and 15.8 emu g^{-1}, respectively. The reduced magnetization originates from the dilution effect of the non-magnetic carbon layer and charge transfer/spin interactions [39–41]. In contrast, CNT-$NC_1{}^M$-NC_1 and CNT-$NC_1{}^M$-NC_2 exhibit higher coercivity (H_c: 156.6 and 132.9 Oe) than CNT-$NC_1{}^M$ (100.8 Oe) (Fig. S4), attributed to weakened magnetic dipole interactions and enhanced interfacial pinning [42,43]. Thermogravimetric analysis (TGA) in air was used to estimate FeCoNi loading in CNT–$NC_1{}^M$–NC_1; however, XRD identified the product as an Fe/Co/Ni ferrite solid solution of unresolvable composition (Fig. S5). Therefore, inductively coupled plasma optical emission spectrometry was employed to quantify the loading, revealing Fe > Ni > Co (Fig. S6). This compositional variation, in turn, determines the overall magnetic performance of the composites.

Transmission electron microscope (TEM) and high-resolution TEM (HRTEM) images of the CNT-$NC_1{}^M$-NC_1 composite are displayed in Figs. 2(e_1–e_6) and S7. As observed, FeCoNi nanoparticles remain well-dispersed without agglomeration (Fig. 2(e_1 and e_2)). HRTEM images (Figs. 2(e_3) and S7(c)) reveal a core-shell FeCoNi nanoparticle/graphitic carbon heterojunction with distinct lattice ordering. In contrast to the metal-catalyzed formation of graphitic carbon featuring clear lattice fringes, the outer PI assemblies yield amorphous carbon without such ordering (Fig. S7(d)). The resulting FeCoNi/graphitic carbon and $NC_1{}^M$/NC_1 interfaces are essential for enhancing interfacial polarization. Similar heterointerfaces (e.g.,

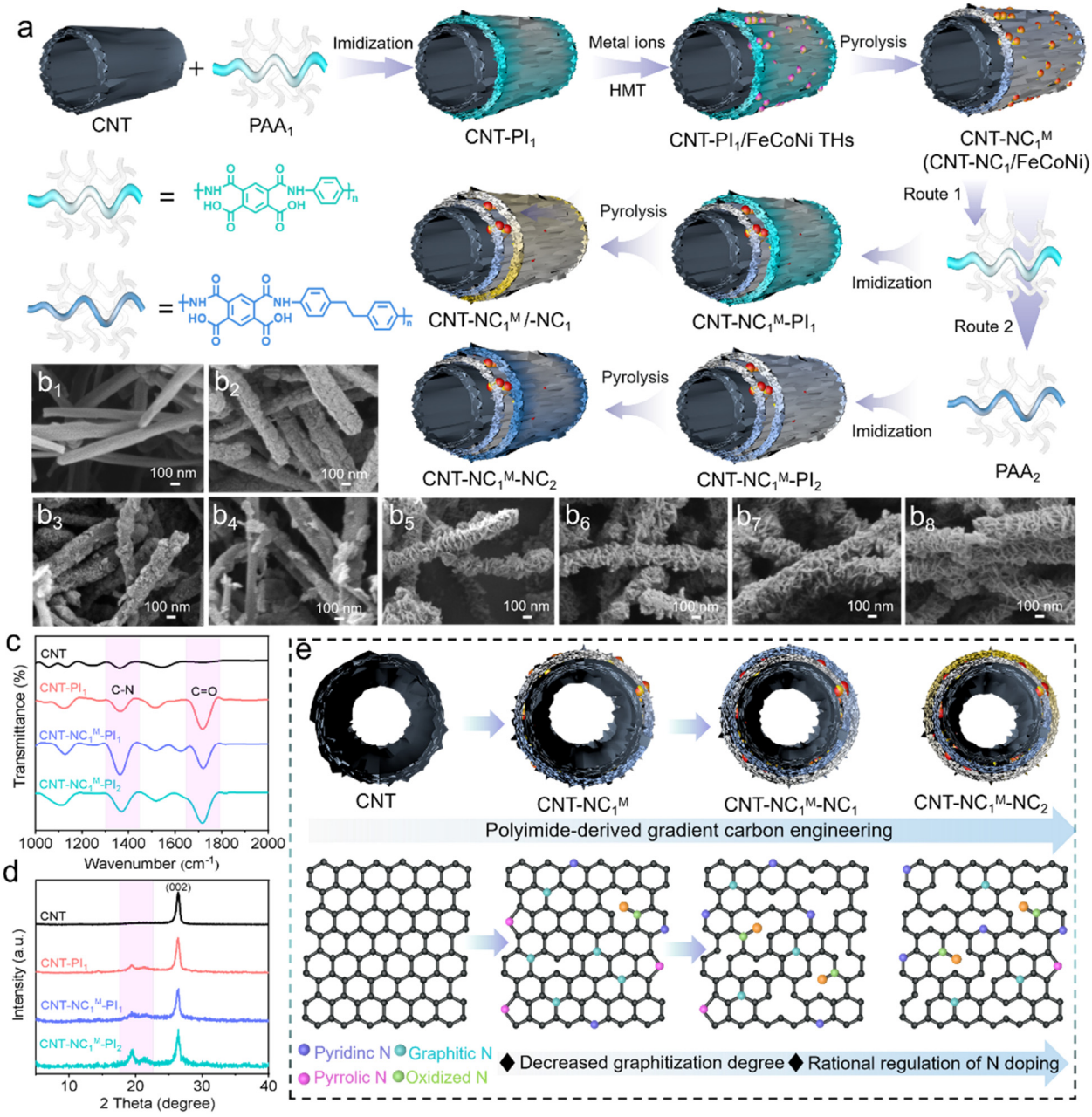


Fig. 1. (a) Schematic fabrication of the gradient carbon nanostructures. (b_1–b_8) SEM images of CNT, CNT-PI_1, CNT-PI_1/FeCoNi THs, CNT-$NC_1{}^M$, CNT-$NC_1{}^M$-PI_1, CNT-$NC_1{}^M$-NC_1, CNT-$NC_1{}^M$-PI_2, and CNT-$NC_1{}^M$-NC_2. (c) FT-IR spectra and (d) XRD patterns of the PI-containing intermediates and CNT. (e) Schematic illustration of the gradient N-doped carbon nanostructures.

FeCoNi/graphitic carbon and Co/graphitic carbon) have also been engineered to promote interfacial polarization [19,39]. Fig. 2(e_4) shows a lattice fringe spacing of 0.35 nm, corresponding to the (002) plane of graphitic carbon, while the 0.20 nm spacing in Fig. 2(e_5) is indexed to the (111) plane of FeCoNi alloy. Localized discontinuities in these fringes indicate defective structures (e.g., point defects) that facilitate polarization relaxation [44]. Graphitic domains (Fig. 2(e_6)) in the carbon matrix further enhance interfacial polarization through additional interfaces and promote conduction loss via ordered microstructures. To verify the gradient nanostructure of CNT–$NC_1{}^M$–NC_1, elemental line profiles are provided in Fig. S8. The profiles show that the carbon peak sharply increases in the inner region but remains low in the outer region, confirming a thickness gradient of the carbon layers. Additionally, metal elements are mainly concentrated in the inner region. The energy-dispersive spectroscopy (EDS) mapping images (Figs. 2(f_1–f_6) and S9) reveal the co-localization of Fe, Co, and Ni elements, confirming the formation of a FeCoNi alloy, while the uniform distribution of C, N, and O signifies homogeneous doping within the carbon matrix.

X-ray photoelectron spectroscopy (XPS) was used to analyze the surface composition and valence states. The survey spectra (Fig. S10(a)) confirm the presence of C, N, O, Fe, Co, and Ni. The high-resolution C 1s spectra (Fig. S10(b)) are deconvoluted into four peaks at 284.8 (C–C/C=C), 285.6 (C–N), 286.5 (C–O), and 289.8 eV (O–C=O) [9,45]. The Fe 2p spectrum of CNT-$NC_1{}^M$ (Fig. 2(g)) reveals metallic Fe (707.3/720.1 eV), Fe^{2+}/Fe^{3+} doublets (710.7/723.7 and 712.9/727.4 eV), and satellite peaks at 716.0/732.6 eV [46]. The Co 2p and Ni 2p spectra (Fig. 2(h and i)) exhibit metallic peaks (778.5/794.9 eV for Co; 853.2/870.8 eV for Ni), oxidized species (Co^{2+} at 781.8/797.2 eV; Ni^{2+} at 854.9/872.8 eV), and satellite peaks (787.3/803.9 and 860.8/878.9 eV). Notably, the intensities

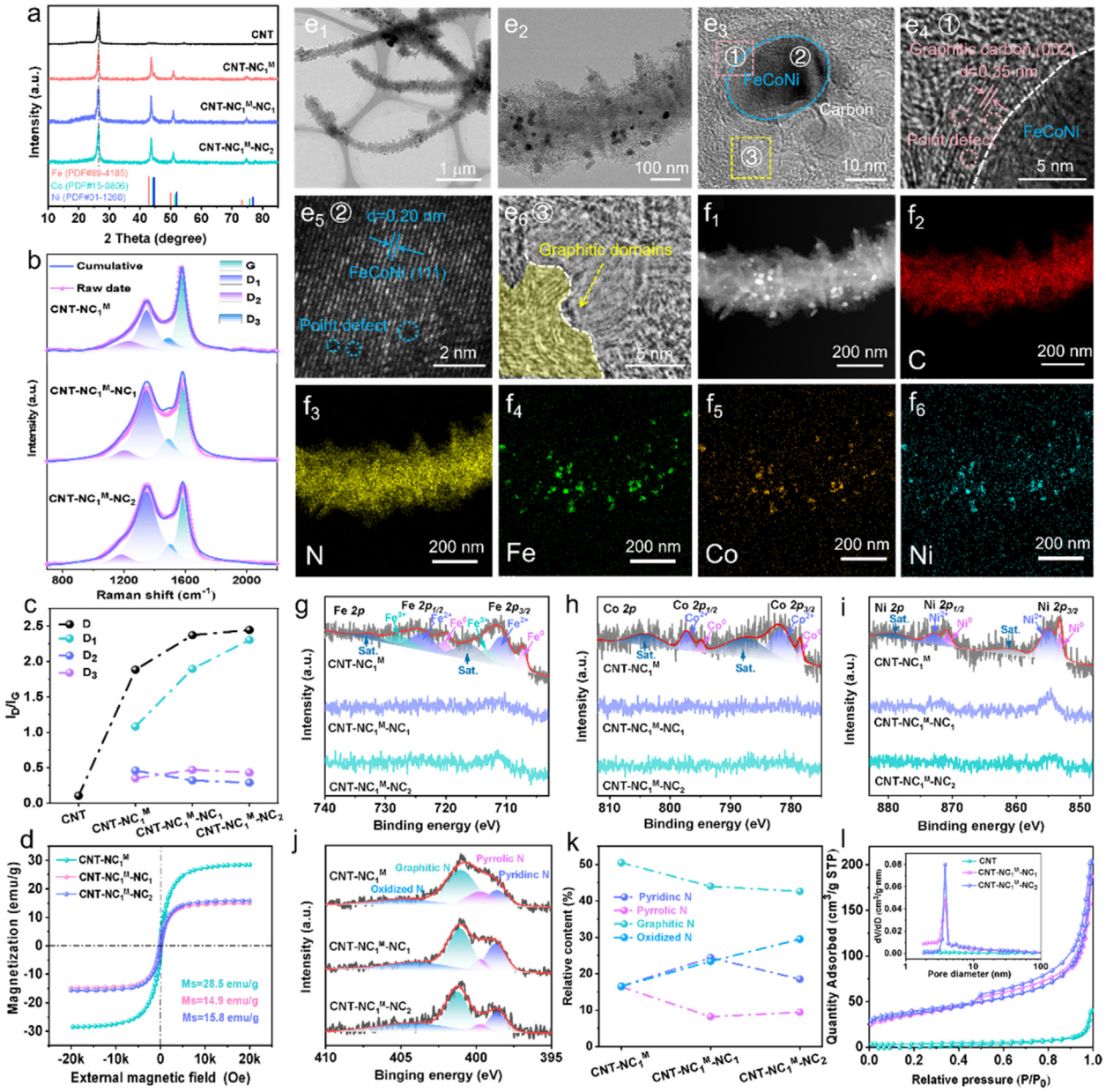


Fig. 2. (a) XRD patterns, (b) Raman spectra deconvoluted into four peaks using Lorentzian fitting. (c) I_D/I_G ratio from two-peak fitting and I_{D1}/I_G, I_{D2}/I_G, I_{D3}/I_G ratios from four-peak fitting. (d) Hysteresis loops. (e and f) TEM, HRTEM, and EDS mapping images of CNT-NC_1^M-NC_1. (g–j) High-resolution Fe 2p, Co 2p, Ni 2p, and N 1s spectra. (k) Relative content of N-doped carbon species. (l) N_2 adsorption-desorption isotherms and pore size distribution.

of all metal peaks are markedly reduced for CNT-NC_1^M-NC_1 and CNT-NC_1^M-NC_2, confirming the successful encapsulation of CNT-NC_1^M by the outer NC_1 and NC_2 layers. High-temperature pyrolysis enables nitrogen doping into graphitic carbons, which can modulate conduction loss. As shown in Fig. 2(j), the high-resolution N 1s spectra are deconvoluted into four peaks assigned to pyridinic N (398.6 eV), pyrrolic N (399.6 eV), graphitic N (401.0 eV), and oxidized N (404.5 eV) [47,48]. Compared with CNT-NC_1^M, both CNT-NC_1^M-NC_1 and CNT-NC_1^M-NC_2 exhibit lower graphitic N and pyrrolic N contents but higher pyridinic N and oxidized N contents (Fig. 2(k)). Specifically, the change in pyridinic N and pyrrolic N contributed to dipolar relaxation, while the decrease in graphitic N diminished conduction loss [49,50]. The high-resolution O 1s spectra (Fig. S11) exhibit peaks at 530.3, 532.1, and 533.6 eV, assigned to Fe–O, C=O, and C–O, respectively. These results confirm both the outer NC_1/NC_2 coating on CNT-NC_1^M and the presence of polar groups in the carbon matrix, which facilitate polarization relaxation. The specific surface area and pore size distribution were determined by N_2 adsorption-desorption measurements (Fig. 2(l)). Unlike CNT-NC_1^M, both CNT-NC_1^M-NC_1 and CNT-NC_1^M-NC_2 exhibit typical Type IV isotherms and possess a mesoporous structure with pore diameters of 3.0–4.0 nm. Brunauer-Emmett-Teller analysis reveals specific surface areas of 129.02 and 142.25 $m^2\ g^{-1}$ for CNT-NC_1^M-NC_1 and CNT-NC_1^M-NC_2, respectively, significantly higher than that of pristine CNT (10.79 $m^2\ g^{-1}$). This porous gradient structure shows significant promise for optimizing complex permittivity and impedance matching [51,52].

2.2. EMW absorption performance and electromagnetic parameters

EMW absorption performance was evaluated by reflection loss (RL) and EAB (defined as the frequency range for RL ≤ −10 dB).

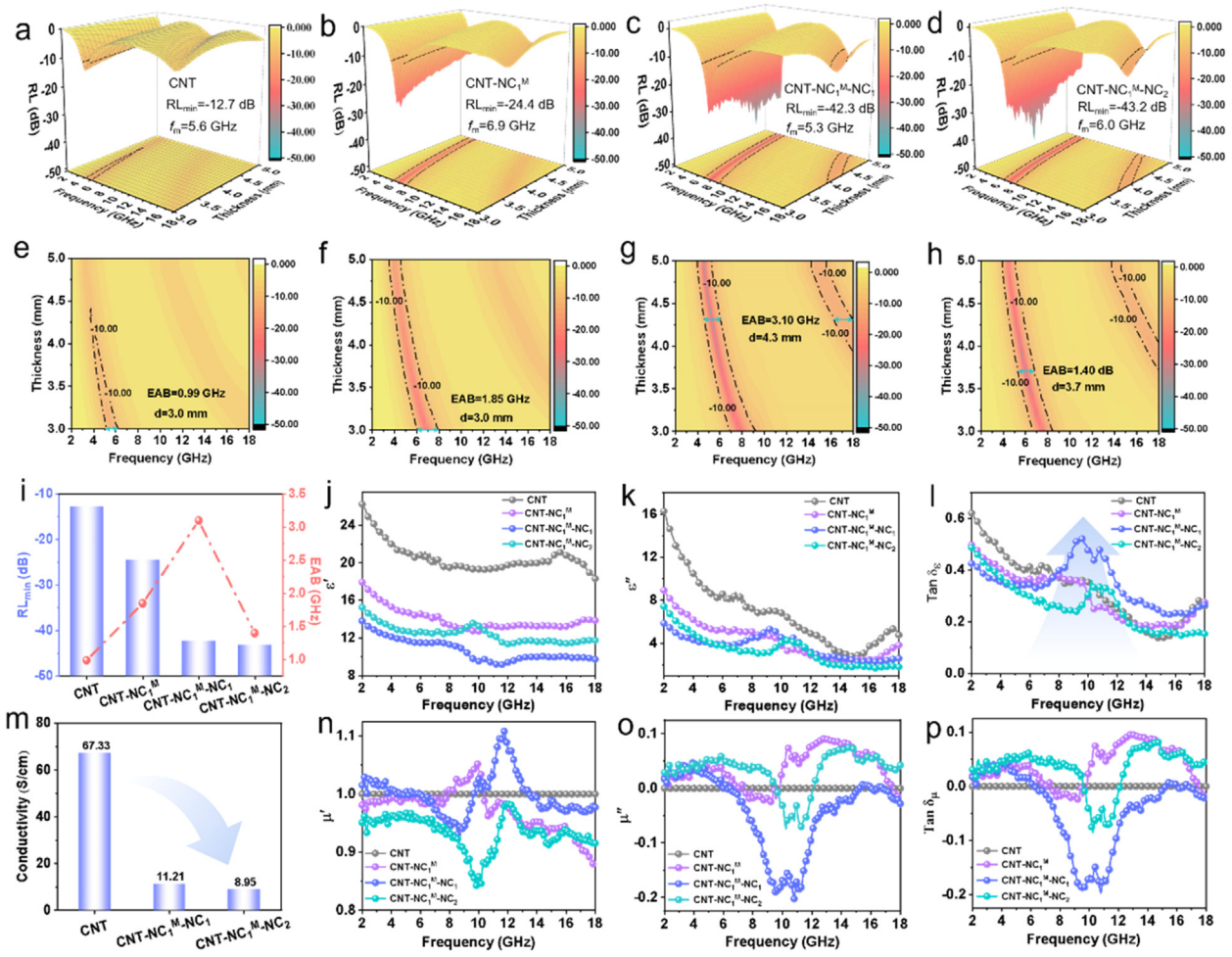


Fig. 3. 3D and 2D RL contour maps: (a and e) CNT, (b and f) CNT-$NC_1{}^M$, (c and g) CNT-$NC_1{}^M$-NC_1, and (d and h) CNT-$NC_1{}^M$-NC_2. (i) Comparison of RL_{min} and EAB. Complex permittivity: (j) real part, (k) imaginary part, and (l) dielectric loss tangent. (m) Electrical conductivity. Complex permeability: (n) real part, (o) imaginary part, and (p) magnetic loss tangent.

Lower RL and larger EAB indicate superior EMW absorption. According to transmission line theory, RL was calculated using Eqs. (1 and 2) [53,54].

$$\mathrm{RL}\ (\mathrm{dB}) = 20\log\left|\frac{Z_{in}-Z_0}{Z_{in}+Z_0}\right| \quad (1)$$

$$Z_{in} = Z_0\sqrt{\mu_r/\varepsilon_r}\tanh\left[j\frac{2\pi fd}{c}\sqrt{\mu_r\varepsilon_r}\right] \quad (2)$$

where Z_0, Z_{in}, ε_r, μ_r, f, d, and c correspond to the free space impedance, normalized input impedance, complex permittivity, complex permeability, frequency, thickness, and speed of light, respectively. As shown in Fig. 3(a and e), CNT exhibits a RL_{min} of −12.7 dB at 5.6 GHz with an EAB of 0.99 GHz. After coating with $NC_1{}^M$ (Fig. 3(b and f)), CNT-$NC_1{}^M$ achieves a RL_{min} of −24.4 dB at 6.9 GHz and a broader EAB of 1.85 GHz (6.08–7.93 GHz). Additional deposition of outer carbon layers (NC_1 and NC_2) further enhances EMW absorption performance. CNT-$NC_1{}^M$-NC_1 (Fig. 3(c and g)) delivers an RL_{min} of −42.3 dB at 5.3 GHz and an EAB of 3.10 GHz, spanning the C band (4.65–6.10 GHz) and Ku band (16.35–18.0 GHz). CNT-$NC_1{}^M$-NC_2 (Fig. 3(d and h)) achieves a RL_{min} of −43.2 dB at 6.0 GHz and an EAB of 1.40 GHz (C band: 4.49–5.89 GHz). A systematic comparison with recently reported literature data covering the C-Ku band is also presented in Table S2. These results demonstrate that gradient carbon engineering progressively enhances EMW absorption (lower RL_{min}, larger EAB, Fig. 3(i)) and yields a dual-band absorption covering both C and Ku bands. Furthermore, the samples with different CNT–$NC_1{}^M$–NC_1 loadings (12wt.% and 15wt.%) also exhibit dual-band EMW absorption, showing tunable EMW absorption properties (Fig. S12).

To elucidate the EMW absorption mechanisms, the complex permittivity ($\varepsilon_r = \varepsilon' - j\varepsilon''$) and permeability ($\mu_r = \mu' - j\mu''$) of all samples were measured using a vector network analyzer over 2–18 GHz. The real parts (ε', μ') represent energy storage, while the imaginary parts (ε'', μ'') correspond to energy dissipation [55–57]. The dielectric and magnetic loss tangents ($\tan\delta_\varepsilon = \varepsilon''/\varepsilon'$ and $\tan\delta_\mu = \mu''/\mu'$) are subsequently calculated to quantify the respective loss extents. CNT exhibits the highest complex permittivity (ε' = 26.3–18.3, ε'' = 16.3–2.9) owing to high electrical conductivity, which, however, causes impedance mismatch and poor EMW absorption (Fig. 3(j and k)). Gradient carbon engineering significantly reduces both ε' and ε''. For CNT-$NC_1{}^M$, CNT-$NC_1{}^M$-NC_1, and CNT-$NC_1{}^M$-NC_2, ε' decreases from 17.9 to 12.6, 13.8 to 9.1, and 15.3 to 11.3, respectively, while ε'' decreases from 8.9 to 2.3, 5.7 to 2.3, and 7.4 to 1.7. According to free-electron theory, $\varepsilon'' = \sigma/2\pi\varepsilon_0 f$ [58,59]. On this basis, gradient carbon engineering progressively reduces σ, thereby lowering dielectric loss. Likewise, the accompanying decrease in ε' reflects a weakened polarization capability. Four–point probe measurements of CNT–$NC_1{}^M$–NC_1 and CNT–$NC_1{}^M$–NC_2 confirm that gradient carbon engineering effectively suppresses the excessive σ of CNT (Fig. 3(m)). Both ε' and ε'' decrease with frequency due to polarization lag, showing typ-

ical frequency dispersion [60]. The tanδ_ε curves (Fig. 3(l)) exhibit multiple relaxation peaks at 8.0–12.0 GHz, indicative of interfacial and dipole polarizations. CNT–NC$_1$^{M}–NC$_1$ shows the highest tanδ_ε in the mid–to–high frequency range, implying the strongest polarization relaxation.

With regard to magnetic response, the significant fluctuations are observed in Fig. 3(n–p), indicating storage and dissipation of magnetic energy under an alternating EM field. Specifically, negative imaginary permeability implies magnetic energy radiation and conversion to electric energy, increasing ε'' for dielectric loss in the corresponding range [61]. Figs. S12 and S13 present the EM parameters and absorption performance of CNT-NC$_1$^{M}-NC$_1$ at 12wt.% and 15wt.% loadings, both showing tunable dual–band absorption. Higher filler loading leads to larger ε' and ε'' values. However, the 12wt.% sample shows higher μ', μ'', and tanδ_μ in a certain frequency range, indicating enhanced magnetic energy storage and dissipation. Meanwhile, the higher-loading sample exhibits generally higher tanδ_ε values, suggesting stronger dielectric loss. To further analyze the magnetic loss mechanism, the eddy current coefficient ($C_0 = \mu''(\mu')^{-2}f^{-1}$) is calculated to evaluate the contributions of eddy current loss, natural resonance, and exchange resonance [62]. Constant C_0 with frequency implies dominant eddy current loss, while significant variation suggests natural/exchange resonance dominance. Fig. S14 exhibits noticeable C_0 fluctuations over the whole frequency range, indicating the presence of natural and exchange resonances. However, these resonances are not dominant for EMW absorption, as tanδ_μ < tanδ_ε for all samples, indicating that dielectric loss prevails. Notably, magnetic loss is also affected by FeCoNi composition heterogeneity. For the three reported samples, the respective weight ratios of Fe, Ni, and Co are (0.5%, 11.0%, and 47.3%), (0.3%, 6.5%, and 44.9%), and (0.4%, 13.9%, and 50.6%), respectively [9]. Higher Ni and Co contents correlate with more pronounced fluctuation peaks in the μ''–f, tanδ_μ–f, and C_0–f curves, indicating stronger magnetic loss from exchange resonance and natural resonance.

The EMW absorption of all samples is further evaluated using impedance matching ($|Z_{in}/Z_0|$) and attenuation constant (α), derived from Eqs. (3 and 4), which quantify EMW penetration and attenuation, respectively [63]. Fig. 4(a–d) shows the relationship among impedance matching, thickness, and frequency. Within the $|Z_{in}/Z_0|$ range of 0.8–1.2 (dashed line, good impedance matching), the area ratios for CNT, CNT-NC$_1$^{M}, CNT-NC$_1$^{M}-NC$_1$, and CNT-NC$_1$^{M}-NC$_2$ are 2.4%, 10.1%, 12.3%, and 11.6%, respectively. Gradient carbon engineering broadens the frequency and thickness ranges for tuning $|Z_{in}/Z_0|$ by reducing electrical conductivity, thereby mitigating the dense conductive network and enabling balanced attenuation with impedance matching. Consequently, the area ratios from the impedance matching maps show an increasing trend. Figs. 4(e) and S15 correlate RL at specific thicknesses with quarter–wavelength ($\lambda/4$) matching thickness ($t_m = n\lambda/4$, n = 1, 3, 5…) and frequency–dependent impedance matching. For CNT-NC$_1$^{M}-NC$_1$ and CNT-NC$_1$^{M}-NC$_2$, $|Z_{in}/Z_0|$ approaches 1 at specific thicknesses, exactly matching the $\lambda/4$ model predictions, indicating that reduced RL arises from good impedance matching, consistent with $\lambda/4$ theory [64]. In contrast, CNT and CNT–NC$_1$^{M} show $|Z_{in}/Z_0|$ deviating further from 1 at corresponding $\lambda/4$ thicknesses, causing poor matching and inferior EMW dissipation (Fig. 4(f)). The attenuation constant α increases with frequency (Fig. 4(g)), implying stronger high–frequency attenuation; however, this does not translate to RL_{min} due to poor impedance matching limiting EMW penetration. Moreover, gradient carbon engineering reduces α across the entire frequency range, suggesting a general decline in attenuation capacity. Thus, the gradient nanostructure design that balances attenuation capability and impedance matching is key to dual-band EMW absorption and EAB broadening. For CNT-NC$_1$^{M}-NC$_1$, low $|Z_{in}/Z_0|$ (0.52–0.63) combined with high α (122.52–136.43) over 16.35–18.0 GHz enables effective absorption (Fig. S16(a)), whereas CNT-NC$_1$^{M}-NC$_2$ fails due to lower α (Fig. S16(b)). Hence, a larger α can compensate for poor impedance matching, irrespective of frequency dependence.

$$|Z_{in}/Z_0| = \sqrt{\mu_r/\varepsilon_r}\tanh\left[j\frac{2\pi fd}{c}\sqrt{\mu_r\varepsilon_r}\right] \quad (3)$$

$$\alpha = \frac{\sqrt{2}\pi f}{c}\sqrt{(\mu''\varepsilon'' - \mu'\varepsilon') + \sqrt{(\mu''\varepsilon'' - \mu'\varepsilon')^2 + (\mu'\varepsilon'' + \mu''\varepsilon')^2}} \quad (4)$$

To better understand the contributions of polarization relaxation and conduction loss to dielectric loss, Cole-Cole curves of all samples were plotted using Eq. (5) [65–68]. In this analysis, Cole-Cole semicircles indicate polarization relaxation (number and radius reflect relaxation capacity), while the tail slope (k) reflects the extent of conduction loss. As shown in Fig. 4(h), distinct irregular semicircles are observed in the Cole-Cole curves of CNT-NC$_1$^{M}-NC$_1$ and CNT-NC$_1$^{M}-NC$_2$ in the low-ε' region, suggesting the enhanced polarization relaxation. In contrast, the linear tails appear in the high-ε' region, which is indicative of conduction loss. Specifically, the k value decreases from 1.43 for CNT to 1.11 (CNT-NC$_1$^{M}), 1.15 (CNT-NC$_1$^{M}-NC$_1$), and 1.20 (CNT-NC$_1$^{M}-NC$_2$), consistent with the variation in σ and indicating reduced conduction loss [44]. Consequently, compared with CNT, gradient carbon engineering effectively balances dielectric loss by enhancing polarization relaxation while suppressing conduction loss.

$$\left(\varepsilon' - \frac{\varepsilon_s + \varepsilon_\infty}{2}\right)^2 + (\varepsilon'')^2 = \left(\frac{\varepsilon_s - \varepsilon_\infty}{2}\right)^2 \quad (5)$$

The excellent EMW absorption of CNT-NC$_1$^{M}-NC$_1$ enables efficient conversion of EM energy into heat, leading to gradual heat accumulation. High thermal conductivity is thus essential for heat transfer. Generally, higher filler loadings enhance thermal conductivity [69–71]. To evaluate the thermal performance, a PDMS-based composite with only 1.8wt.% filler was prepared. Fig. 4(i) shows the thermal conductivity of PDMS, CNT/PDMS, and CNT-NC$_1$^{M}-NC$_1$/PDMS composites. Compared with PDMS (0.158 W m^{-1} K^{-1}), the thermal conductivities of CNT/PDMS and CNT-NC$_1$^{M}-NC$_1$/PDMS increase to 0.226 and 0.175 W m^{-1} K^{-1}, respectively. Notably, the composite retains appreciable thermal conductivity even at such a low loading, implying that higher loadings could further improve heat transfer. A systematic comparison with recent literature on thermal management and EMW absorption is provided in Table S1, indicating the uniqueness of this work. Mechanistically, pristine CNTs possess graphitic carbon structures that facilitate efficient phonon transport, delivering high σ (67.33 S/cm) and thermal conductivity. In contrast, the amorphous carbon layer on CNTs introduces a disordered atomic arrangement, causing strong phonon scattering and low σ (11.21 and 8.95 S/cm). Moreover, it also acts as a thermal barrier, raising interfacial thermal resistance and reducing thermal conduction. Consequently, the gradient carbon nanostructure exhibits lower thermal conductivity than pristine CNTs. To further evaluate thermal conduction, CNT-NC$_1$^{M}-NC$_1$/PDMS and PDMS were placed on a 200 °C hot plate for 10 min, and their surface temperatures were recorded using an infrared thermal camera. Fig. 4(j) presents the temperature rise curves. Compared with PDMS, CNT-NC$_1$^{M}-NC$_1$/PDMS shows a higher surface temperature and faster temperature rise rate, demonstrating rapid heat conduction. Corresponding thermal infrared images are summarized in Fig. 4(k).

2.3. Theoretical calculations and EMW absorption mechanisms

Gradient carbon engineering creates multiple interfaces that facilitate electron transfer and non-uniform charge distribution,

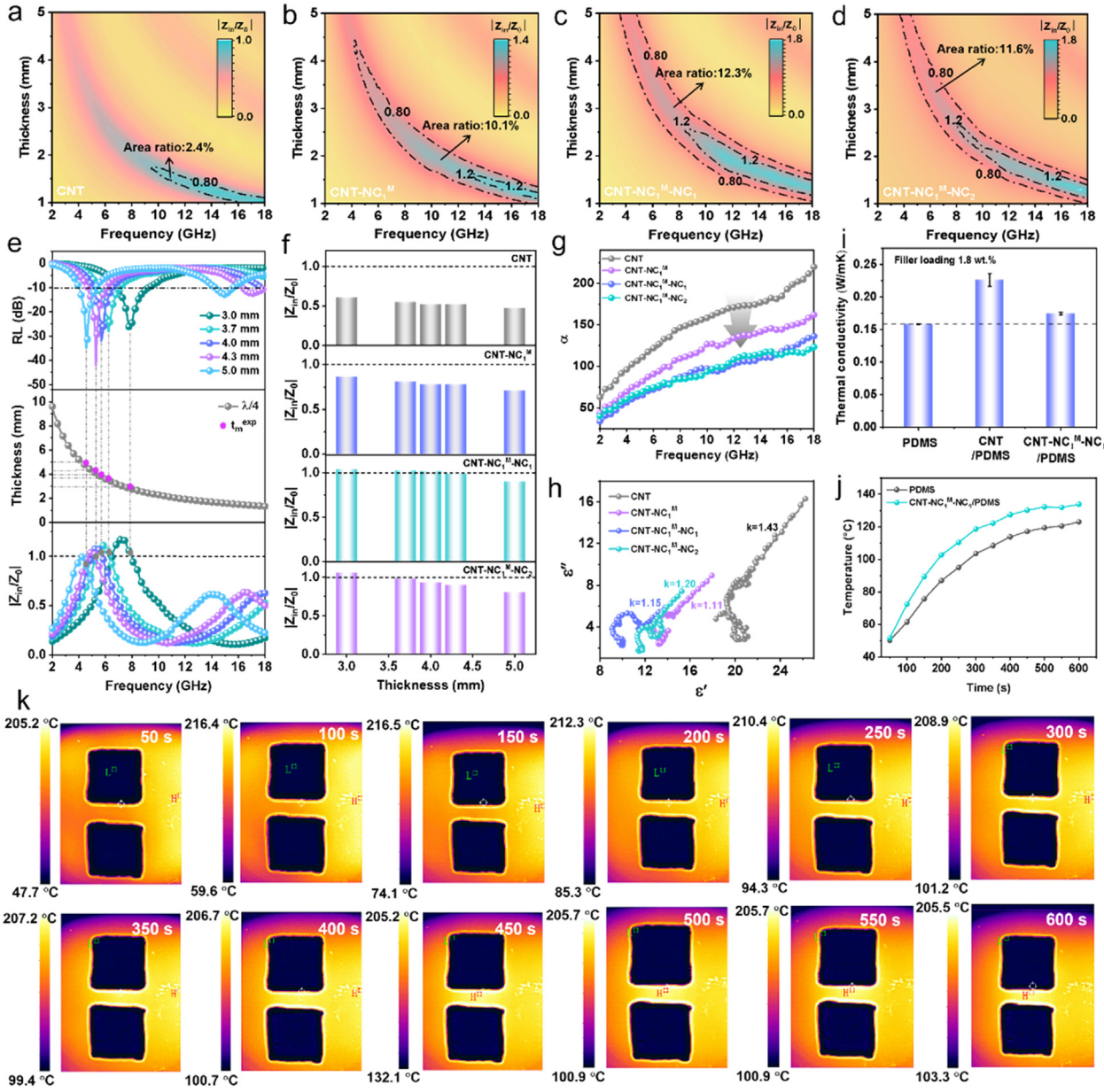


Fig. 4. (a–d) 2D impedance matching maps for all samples. (e) RL curves, quarter-wavelength matching thickness, and frequency-dependent impedance matching for CNT-$NC_1{}^M$-NC_1. (f) Comparison of $|Z_{in}/Z_0|$. (g) Attenuation constant. (h) Cole-Cole curves. (i) Thermal conductivity. (j) Temperature rise curves and (k) thermal infrared images of CNT-$NC_1{}^M$-NC_1/PDMS (top) and PDMS (bottom).

forming BIEF for the enhancement of interfacial polarization [72]. DFT calculations were performed to clarify the role of CNT/$NC_1{}^M$ and $NC_1{}^M$/NC_1 interfaces in CNT–$NC_1{}^M$–NC_1. As shown in Fig. 5(a–c), $NC_1{}^M$ (3.940 eV) has a lower work function compared to CNT (4.595 eV) and NC_1 (4.196 eV). Electrons migrate from $NC_1{}^M$ to both CNT and NC_1 to align the Fermi levels after contact (Fig. 5(d)), inducing band bending at the interfaces and increasing interfacial resistance [43]. Differential charge density analysis reveals electron accumulation on NC_1 (yellow regions) and depletion on $NC_1{}^M$ (cyan regions) at the $NC_1{}^M$/NC_1 interface, thereby establishing a BIEF from $NC_1{}^M$ to NC_1 (Fig. 5(e)). In contrast, the CNT/$NC_1{}^M$ interface shows slight electron accumulation and depletion, indicating a weak BIEF. The initial configurations of CNT, $NC_1{}^M$, and NC_1 are provided in Fig. S17. When the BIEF opposes the applied electric field, charge separation continues until dynamic equilibrium is reached, leading to electron accumulation on the CNT and NC_1 sides [73]. When the applied field aligns with the BIEF, electrons migrate toward $NC_1{}^M$ but are hindered by the BIEF driving force (F_D), establishing a new equilibrium (Fig. 5(f)). Consequently, the alternating electric field induces enhanced interface polarization and thereby modulates EM response.

Benefiting from gradient carbon nanostructures with decreased graphitization degree (Fig. 5(g)), the enhanced EMW absorption is attributed to programmable dielectric loss from gradient carbon engineering and dual-BIEFs (Fig. 5(h)). Key synergistic factors include: (i) Dual-BIEFs amplify interfacial polarization as the main contributor to polarization relaxation [74]; (ii) gradient carbon engineering enhances polarization relaxation and suppresses low-frequency conduction loss for balanced dielectric loss; (iii) gradient nanostructures induce multiple reflections/scattering, prolong-

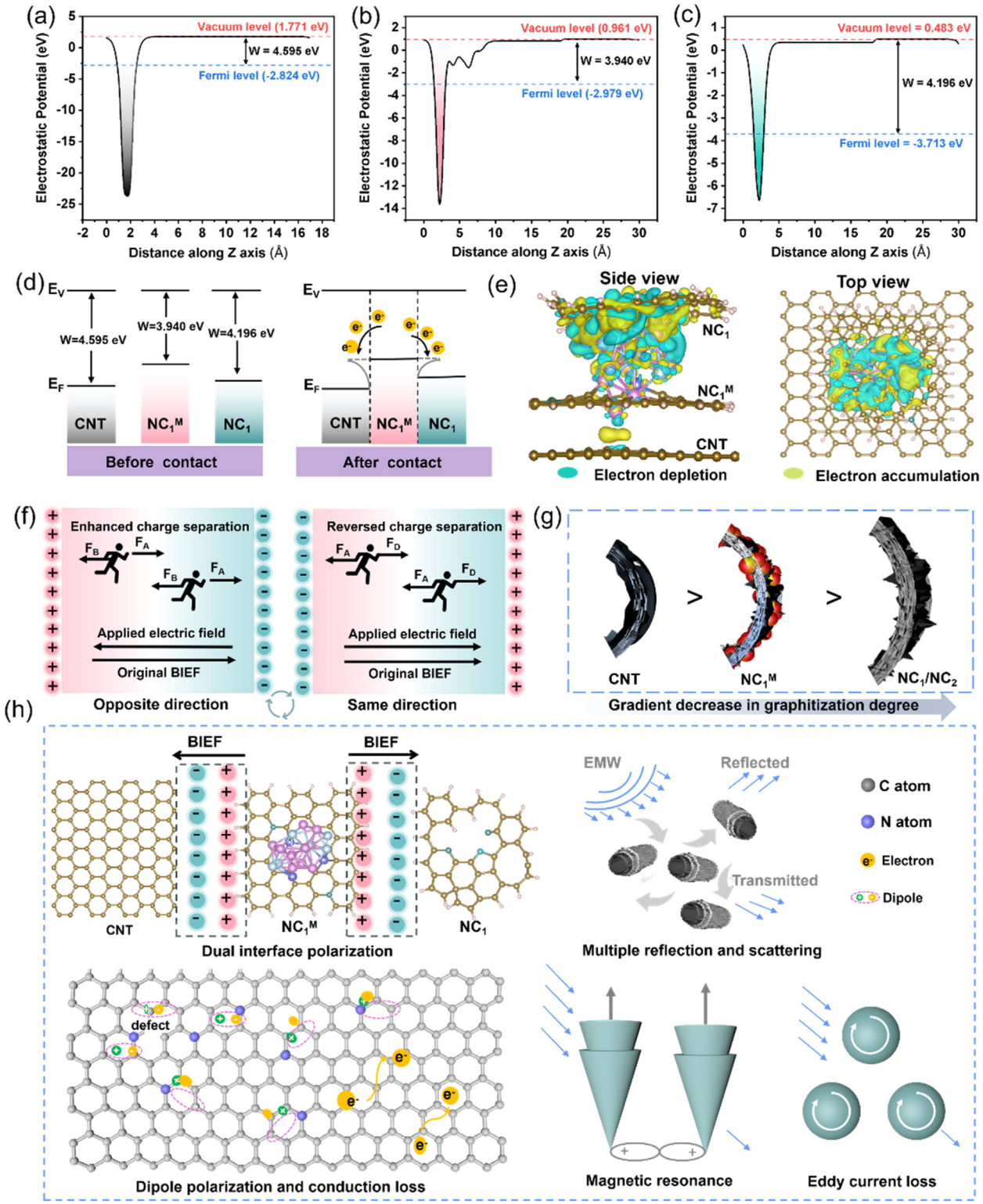


Fig. 5. (a–c) Work function derived from Fermi energy levels. (d) Schematics of energy band alignment and electron redistribution. (e) 3D differential charge density distributions at the CNT/$NC_1{}^M$/NC_1 heterointerfaces. (f) Interfacial charge transport under an alternating electric field. (g) Schematic of carbon layers with varying graphitization degree. (h) Schematics of the EMW absorption mechanisms.

ing EMW paths; (iv) magnetic resonance (natural and exchange resonance) further boosts EMW attenuation [75,76]. Overall, gradient carbon engineering enables dual-BIEFs at the CNT/$NC_1{}^M$ and $NC_1{}^M$/NC_1 interfaces, modulating charge distribution and optimizing EMW absorption performance.

3. Conclusions

In summary, gradient carbon nanostructures with programmable dielectric loss were synthesized via polyimide assembly and pyrolysis. The engineered interfaces and heteroatomic

doping promote dual-BIEF-enhanced interfacial polarization and dipole polarization (from defects/polar groups), significantly boosting polarization relaxation. Concurrently, the low-graphitized and amorphous carbon layers effectively suppress conduction loss by reducing σ. This optimized balance renders dielectric loss dominant over magnetic loss. Leveraging synergistic attenuation and matched impedance, CNT-NC_1^M-NC_1 achieves dual-band EMW absorption with an RL_{min} of −42.3 dB at 5.3 GHz and an EAB of 3.1 GHz across the C and Ku bands. Moreover, the composite also exhibits good thermal conductivity when mixed with PDMS. This work demonstrates the efficacy of gradient carbon engineering in tailoring conduction loss and polarization relaxation for advanced EMW absorption, providing a paradigm for high-performance absorber design through rational dual-BIEF construction.

CRediT authorship contribution statement

Qingfu Ban: Writing – original draft, Funding acquisition, Conceptualization. **Huilin Zhang:** Investigation, Data curation. **Yuejie Song:** Methodology, Investigation. **Jie Liu:** Resources, Funding acquisition. **Yusheng Qin:** Validation, Resources. **Tiantian Zhang:** Methodology, Investigation. **Jie Kong:** Validation, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jmst.2026.06.046.

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